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Computational Logic Handouts

October 23, 2025

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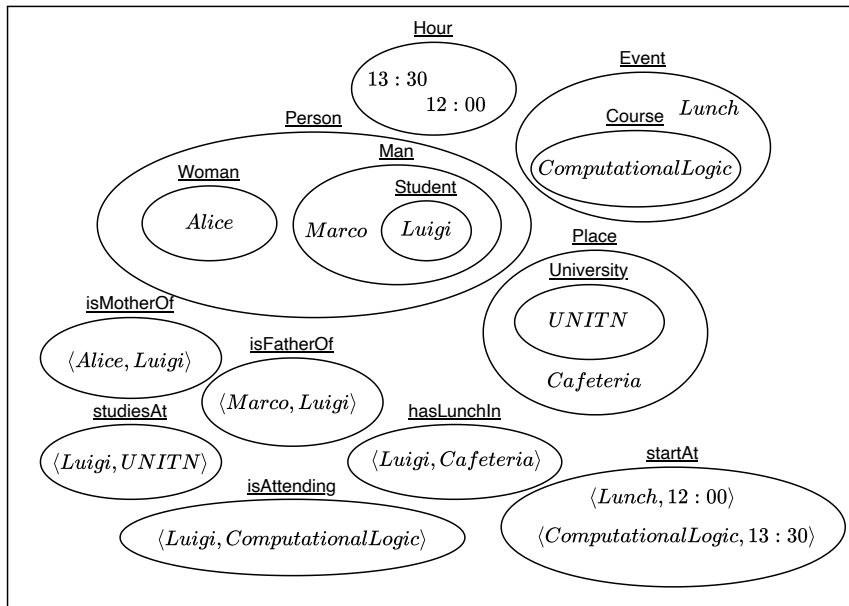
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Chapter 1

LoE

1.1 LoE models, theories and interpretation function

Exercise 1.1 Considering the intended model M represented as a Venn diagram below, determine which of the following pairs of interpretation functions I and theories T are correct with respect to M .



1. $I_e(\text{Marco}) = \text{Marco}$
 $I_e(\text{Luigi}) = \text{Luigi}$
 $I_e(\text{Alice}) = \text{Alice}$
 $I_E(\text{Woman}) = \text{Woman}$
 $I_E(\text{Student}) = \text{Student}$
 $I_E(\text{Man}) = \text{Man}$
 $I_O(\text{motherOf}) = \text{motherOf}$
 $I_O(\text{fatherOf}) = \text{fatherOf}$
- $\text{Man}(\text{Marco})$
 $\text{Student}(\text{Luigi})$
 $\text{Woman}(\text{Alice})$
 $\text{motherOf}(\text{Alice}, \text{Luigi})$
 $\text{fatherOf}(\text{Marco}, \text{Luigi})$

$I_e(\text{Marco}) = \text{Marco}$

$I_e(\text{Luigi}) = \text{Luigi}$

$I_e(\text{Alice}) = \text{Alice}$

2. $I_E(\text{Woman}) = \text{Woman}$

$I_E(\text{Student}) = \text{Student}$

$I_E(\text{Man}) = \text{Man}$

$I_O(\text{motherOf}) = \text{motherOf}$

$I_O(\text{fatherOf}) = \text{fatherOf}$

$\text{Student}(\text{Marco})$

$\text{Man}(\text{Luigi})$

$\text{Woman}(\text{Alice})$

$\text{motherOf}(\text{Alice}, \text{Luigi})$

$\text{fatherOf}(\text{Marco}, \text{Luigi})$

$I_e(\text{Marco}) = \text{Luigi}$

$I_e(\text{Luigi}) = \text{Marco}$

$I_e(\text{Alice}) = \text{Alice}$

3. $I_E(\text{Woman}) = \text{Woman}$

$I_E(\text{Student}) = \text{Student}$

$I_E(\text{Man}) = \text{Man}$

$I_O(\text{motherOf}) = \text{motherOf}$

$I_O(\text{fatherOf}) = \text{fatherOf}$

$\text{Student}(\text{Marco})$

$\text{Man}(\text{Luigi})$

$\text{Woman}(\text{Alice})$

$\text{motherOf}(\text{Alice}, \text{Marco})$

$\text{fatherOf}(\text{Marco}, \text{Marco})$

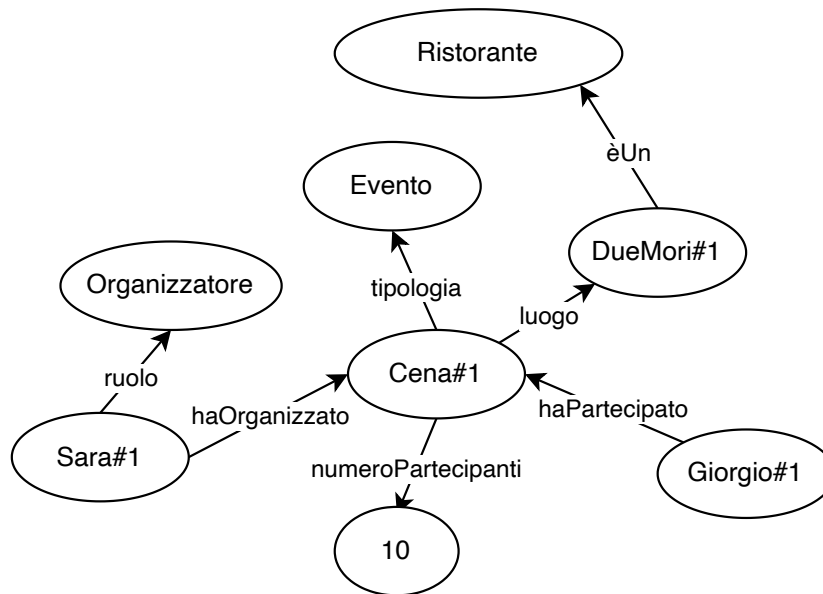
Exercise 1.2 Let there be a table, shown below, showing the grades of an exam (not explicitly stated). Determine if for the domain $D = \langle E, \{C\}, \{R\} \rangle$ of the Logic of Entities below, there is an interpretation function that correctly formalizes the contents of the table. Assume that: (1) there are no synonyms, (2) Mark is formalized as the datatype integer, (3) «entity» is the root etype for all entities, (4) «dtype» is the root etype for all data type values.

Student	Domicile	Mark
MarioRossi	Trento	27
StefaniaBianchi	Napoli	30

1. $E = \text{MarioRossi}\#1, \text{StefaniaBianchi}\#1, \text{Trento}\#1, \text{Naples}\#1, \text{twentyseven}, \text{thirty}$
 $\{C\} = \text{Student, entity, integer, dtype}$
 $\{R\} = \text{Domicile, Mark}$
2. $E = \text{MarioRossi}\#1, \text{StefaniaBianchi}\#1, \text{Trento}\#1, \text{Naples}\#1, 27, 30$
 $\{C\} = \text{Student, city, entity, integer, dtype}$
 $\{R\} = \text{Domicile, Mark}$
3. $E = \text{MarioRossi}, \text{StefaniaBianchi}, \text{Trento}, \text{Naples}, 27, 30$
 $\{C\} = \text{Student, entity, integer, dtype}$
 $\{R\} = \text{Domicile, Mark}$
4. $E = \text{MarioRossi}\#1, \text{StefaniaBianchi}\#1, \text{Trento}\#1, \text{Naples}\#1, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30$
 $\{C\} = \text{Student, entity, integer, dtype}$
 $\{R\} = \text{Domicile, Mark}$
5. $E = \text{MarioRossi}\#1, \text{StefaniaBianchi}\#1, \text{Trento}\#1, 27, 30$
 $\{C\} = \text{Student, integer, entity, dtype}$
 $\{R\} = \text{Domicile, Mark}$
6. $E = \text{MarioRossi}\#1, \text{StefaniaBianchi}\#1, \text{Bolzano}\#1, \text{Rome}\#1, 27, 30$
 $\{C\} = \text{Student, integer, entity, dtype}$
 $\{R\} = \text{Domicile, Mark}$
7. $E = \text{MR}, \text{SB}, \text{TN}, \text{NA}, 27, 30$
 $\{C\} = \text{Student, Domicile, integer, entity, dtype}$
 $\{R\} = \text{Mark}$

8. $E = \text{MarioRossi\#1, StefaniaBianchi\#1, Trento\#1, Naples\#1, 27, 30}$
 $\{C\} = \text{Student, integer, entity, dtype}$

Exercise 1.3 Consider the Knowledge Graph below. Assume there are no synonyms. Assume the following: (1) “numeroPartecipanti” is formalized using the data type integer, (2) “entity” is the entity types of all entities, and (3) dtype is the datatype of all datatype values. Only the names of instances contain the symbol “#”, numbers are datatype values. Indicate for which of the LoE interpretation domains D listed below, with $D = \langle E, \{C\}, \{R\} \rangle$, there exists an interpretation function which correctly formalizes the graph contents.



1. $E = \text{Sara\#1, Giorgio\#1, Cena\#1, DueMori\#1, 10}$
 $\{C\} = \text{Organizzatore, Evento, Ristorante, integer, entity, dtype}$
 $\{R\} = \text{haOrganizzato, haPartecipato, luogo, numeroPartecipanti}$
2. $E = \text{Sara\#1, Giorgio\#1, Cena\#1, DueMori\#1, 10.0}$
 $\{C\} = \text{Organizzatore, Evento, Ristorante, entity, dtype}$
 $\{R\} = \text{haOrganizzato, haPartecipato, luogo, numeroPartecipanti}$
3. $E = \text{Sara\#1, Giorgio\#1, Cena\#1, DueMori\#1, 10}$
 $\{C\} = \text{Organizzatore, Evento, Ristorante, Lavoratore, dtype}$
 $\{R\} = \text{haOrganizzato, haPartecipato, luogo, numeroPartecipanti}$
4. $E = \text{Sara\#1, Giorgio\#1, Cena\#1, DueMori\#1, 10}$
 $\{C\} = \text{Evento, Ristorante, entity, dtype}$
 $\{R\} = \text{haOrganizzato, haPartecipato, luogo, numeroPartecipanti}$
5. $E = \text{Paola, Giorgio, Cena\#1, DueMori\#1, 10}$
 $\{C\} = \text{Organizzatore, Evento, Pizzeria, entity, dtype}$
 $\{R\} = \text{haOrganizzato, haPartecipato, luogo, numeroPartecipanti}$
6. $E = \text{Sara\#1, Giorgio\#1, Cena, DueMori, 10}$
 $\{C\} = \text{Organizzatore, Evento, Ristorante, entity, dtype}$
 $\{R\} = \text{haOrganizzato, haPartecipato, numeroPartecipanti}$
7. $E = \text{Sara, Giorgio, Cena, DueMori, 10, 11, 12, 13, 14.0, 15.1, 10.2}$
 $\{C\} = \text{Organizzatore, Evento, Ristorante, entity, integer, real, dtype}$
 $\{R\} = \text{haOrganizzato, haPartecipato, luogo, numeroPartecipanti}$

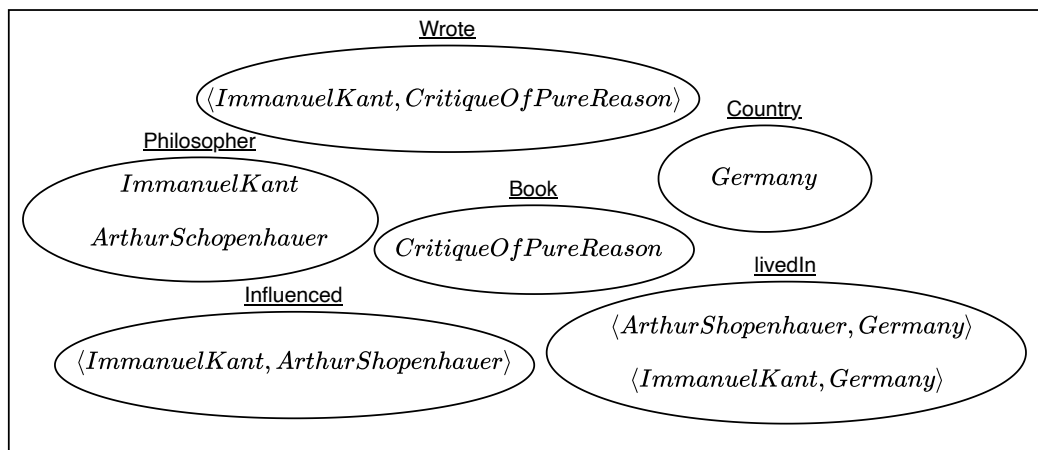
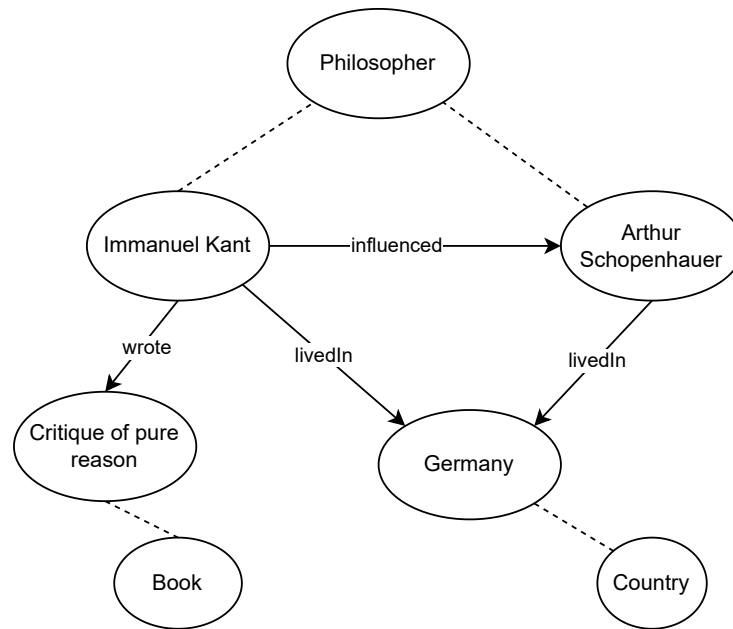
Exercise 1.4 Consider the table below. Assume there are no synonyms. Assume the following: (1) “anno di stampa” and “data prestito” are formalized using the data type date, (2) “entity” is the entity types of all entities, and (3) dtype is the datatype of all datatype values. Indicate for which of the LoE interpretation domains D listed below, with $D = \langle E, \{C\}, \{R\} \rangle$, there exists an interpretation function which correctly formalizes the graph contents.

Libro	Anno di stampa	Prestato a	Data prestito
Essential Software Architecture	2011	Giovanni Rossi	2024/10/12
Compilatori	2019	Paola Merini	2025/01/08
Algoritmi e strutture dati	2016	Stefano Porti	2024/12/21

1. $E = \text{EssentialSoftwareArchitecture\#1, Compilatori\#1, AlgoritmiEStruttureDati\#1, GiovanniRossi\#1, PaolaMerini\#1, StefanoPorti\#1, 2011, 2019, 2016, 2024/10/12, 2025/01/08, 2024/12/21}$
 $\{C\} = \text{Libro, Persona, integer, date, entity, dtype}$
 $\{R\} = \text{annoStampa, prestatoA, prestatoInData}$
2. $E = \text{ESA\#1, C\#1, Asd\#1, GR\#1, PM\#1, SPi\#1, 2011, 2019, 2016, 2024/10/12, 2025/01/08, 2024/12/21}$
 $\{C\} = \text{Lib, Prs, integer, date, entity, dtype}$
 $\{R\} = \text{aS, pA, pD}$
3. $E = \text{EssentialSoftwareArchitecture\#1, Compilatori\#1, AlgoritmiEStruttureDati\#1, GiovanniRossi\#1, PaolaMerini\#1, StefanoPorti\#1, 2011, 2019, 2016, 2024/10/12, 2025/01/08, 2024/12/21}$
 $\{C\} = \text{Libro, PrestatoA, integer, date, entity, dtype}$
 $\{R\} = \text{annoStampa, prestatoInData}$
4. $E = \text{EssentialSoftwareArchitecture\#1, Compilatori\#1, AlgoritmiEStruttureDati\#1, GiovanniRossi\#1, PaolaMerini\#1, StefanoPorti\#1, 2011, 2012, 2013, 2014, 2015, 2016, 2019, 2016, 2024/10/12, 2025/01/08, 2024/12/21}$
 $\{C\} = \text{integer, date, entity, dtype}$
 $\{R\} = \text{annoStampa, prestatoA, prestatoInData}$
5. $E = \text{EssentialSoftwareArchitecture\#1, Compilatori\#1, AlgoritmiEStruttureDati\#1, GiovanniRossi\#1, PaolaMerini\#1, StefanoPorti\#1, 2011, 2019, 2016, 2024/11/12, 2025/01/08, 2024/12/21}$
 $\{C\} = \text{Libro, Persona, integer, date, entity, dtype}$
 $\{R\} = \text{anno, prestato, inData}$
6. $E = \text{ESA\#1, C\#1, Asd\#1, GR\#1, PM\#1, SPi\#1, 2011, 2019, 2016, 2024/10/12, 2025/01/08, 2024/12/21}$
 $\{C\} = \text{Libro, Persona, integer, date}$
 $\{R\} = \text{annoStampa, prestatoA, prestatoInData}$

1.2 Translation across formalization levels

Exercise 1.5 Given the following graph as semi-formal representation, and the corresponding intended model, determine which of the presented pairs of interpretation functions and theories are a correct formalization of the graph.

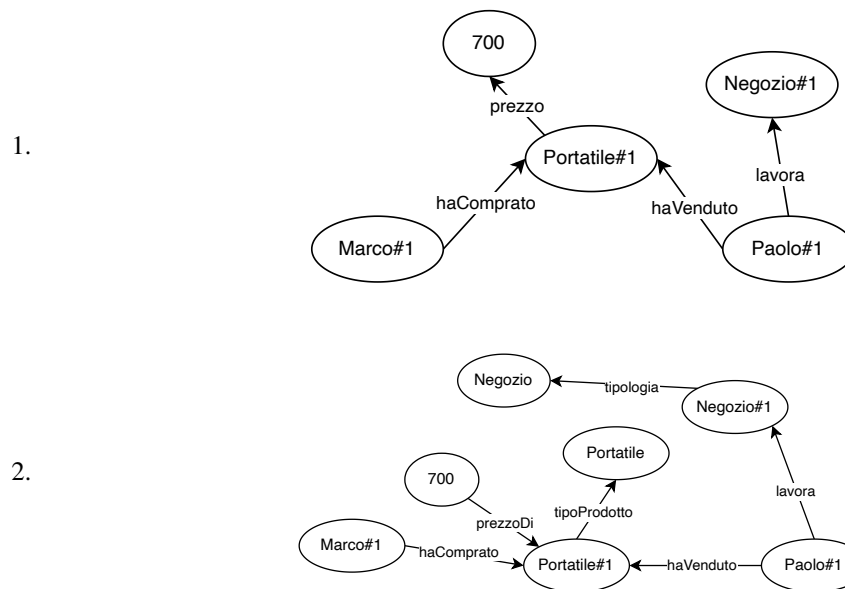


- $I_e(Kant) = ImmanuelKant$
 $I_e(Schopenhauer) = ArthurSchopenhauer$
 $I_e(Germany) = Germany$
 $I_e(Critique) = CritiqueOfPureReason$
 $I_E(Book) = Book$
 $I_E(Country) = Country$
 $I_E(Philosopher) = Philosopher$
 $I_O(wrote) = wrote$
 $I_O(influenced) = influenced$
 $I_O(livedIn) = livedIn$

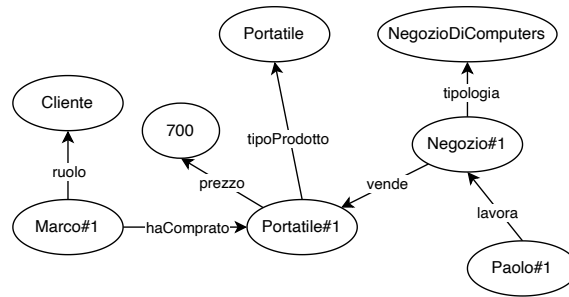
$Philosopher(Kant)$
 $Philosopher(Schopenhauer)$
 $Country(Germany)$
 $Book(Critique)$
 $livedIn(Schopenhauer, Germany)$
 $livedIn(Kant, Germany)$
 $influenced(Kant, Schopenhauer)$
 $wrote(Kant, Critique)$

- $I_e(Kant) = ImmanuelKant$
 $I_e(Schopenhauer) = ArthurSchopenhauer$
 $I_e(Germany) = Germany$
 $I_e(Critique) = CritiqueOfPureReason$
 $I_E(Book) = Book$
 $I_E(Country) = Country$
 $I_E(Philosopher) = Philosopher$
 $I_O(wrote) = wrote$
 $I_O(influenced) = influenced$
 $I_O(livedIn) = livedIn$
- $I_e(Kant) = ImmanuelKant$
 $I_e(Schopenhauer) = ArthurSchopenhauer$
 $I_e(Germany) = Germany$
 $I_e(Critique) = CritiqueOfPureReason$
 $I_E(Book) = Book$
 $I_E(Country) = Country$
 $I_E(Philosopher) = Philosopher$
 $I_O(hasAuthor) = wrote$
 $I_O(influenced) = influenced$
 $I_O(livedIn) = livedIn$
- $I_e(K) = ImmanuelKant$
 $I_e(S) = ArthurSchopenhauer$
 $I_e(G) = Germany$
 $I_e(C) = CritiqueOfPureReason$
 $I_E(Book) = Book$
 $I_E(Country) = Country$
 $I_E(Philosopher) = Philosopher$
 $I_O(hasAuthor) = wrote$
 $I_O(influenced) = influenced$
 $I_O(livedIn) = livedIn$
- $Philosopher(Kant)$
 $Philosopher(Schopenhauer)$
 $Country(Germany)$
 $Book(Critique)$
 $livedIn(Schopenhauer, Germany)$
 $livedIn(Kant, Germany)$
 $influenced(Kant, Schopenhauer)$
 $authorOf(Kant, Critique)$
- $Philosopher(Kant)$
 $Philosopher(Schopenhauer)$
 $Country(Germany)$
 $Book(Critique)$
 $livedIn(Schopenhauer, Germany)$
 $livedIn(Kant, Germany)$
 $influenced(Kant, Schopenhauer)$
 $hasAuthor(Kant, Critique)$
- $Philosopher(K)$
 $Philosopher(S)$
 $Country(G)$
 $Book(C)$
 $livedIn(S, G)$
 $livedIn(K, G)$
 $influenced(K, S)$
 $hasAuthor(C, K)$

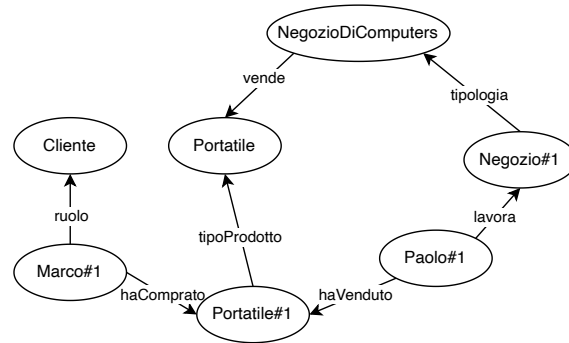
Exercise 1.6 Determine which of following informal knowledge graphs can be fully formalized in LoE, without loss of information. Only the names of instances contain the symbol “#”; numbers must be considered as values of dtypes.



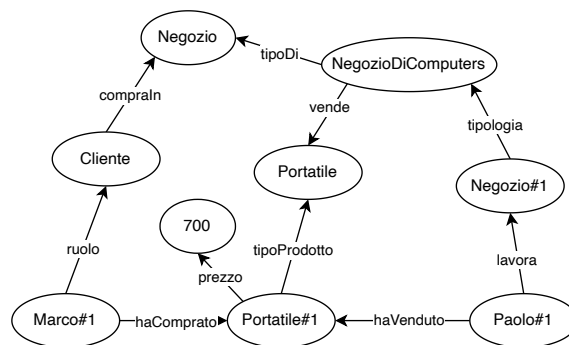
3.



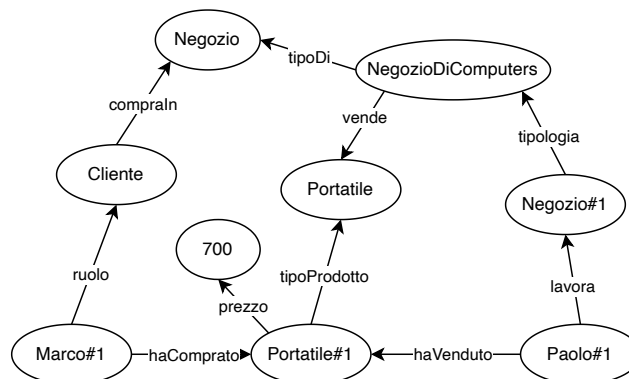
4.



5.



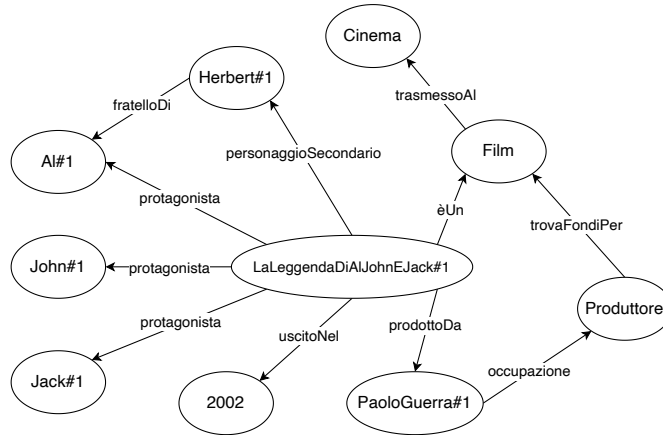
Exercise 1.7 Consider the knowledge graph below. Only the names of entities contain the symbol “#”, numbers must be considered as values of dtypes. Which of the statements below are true?



1. If the triple $\langle portatile\#1, tipoProdotto, Portatile \rangle$ was formalized in LoE, “*tipoProdotto*” would have to be formalized as an object property.

2. If the triple $\langle \text{portatile}\#1, \text{Prezzo}, 700 \rangle$ was formalized in LoE, “Prezzo” would have to be formalized as a data property.
3. In LoE, the triple $\langle \text{Negozio}\#1, \text{tipologia}, \text{NegozioDiComputers} \rangle$ should be formalized as “Negozio#1” member of the etype “NegozioDiComputers”.
4. The triple $\langle \text{NegozioDiComputers}, \text{vende}, \text{portatile} \rangle$ can be formalized in LoE.
5. The triple $\langle \text{NegozioDiComputers}, \text{tipoDi}, \text{negozio} \rangle$ can be formalized in LoE.

□ Consider the knowledge graph below. Only the names of entities contain the symbol #, numbers must be considered as values of data types. W



1. If the triple $\langle \text{Herbert}\#1, \text{fratelloDi}, \text{Al}\#1 \rangle$ was formalized in LoE, “fratelloDi” would have to be formalized as an object property.
2. If the triple $\langle \text{LaLeggendaDiAlJohnEJack}\#1, \text{uscitoNel}, 2002 \rangle$ was formalized in LoE, “uscitoNel” would have to be formalized as an object property.
3. The triple $\langle \text{PaoloGuerra}\#1, \text{occupazione}, \text{Produttore} \rangle$ can be formalized in LoE.
4. The triple $\langle \text{Film}, \text{trasmessoAl}, \text{Cinema} \rangle$ can be formalized in LoE.
5. In LoE, the entity $\text{Al}\#1$ would be formalized as a member of the etype *protagonista*.

1.3 LoE entailment

Exercise 1.8 Considering the following LoE theory T , indicate which of the following statements are true.

$\text{Person}(\text{Alice})$
 $\text{Person}(\text{Bob})$
 $\text{Person}(\text{Bill})$
 $\text{City}(\text{Paris})$
 $\text{Monument}(\text{EiffelTower})$
 $\text{Museum}(\text{Louvre})$
 $\text{Painting}(\text{MonaLisa})$
 $\text{Painter}(\text{LeonardoDaVinci})$
 $\text{isFriendOf}(\text{Bob}, \text{Alice})$
 $\text{hasVisited}(\text{Bill}, \text{EiffelTower})$
 $\text{hasVisited}(\text{Bob}, \text{Louvre})$
 $\text{hasTraveledTo}(\text{Bob}, \text{Paris})$
 $\text{paintedBy}(\text{MonaLisa}, \text{LeonardoDaVinci})$
 $\text{exhibitedIn}(\text{MonaLisa}, \text{Louvre})$
 $\text{interestedIn}(\text{Bob}, \text{MonaLisa})$
 $\text{isLocatedIn}(\text{EiffelTower}, \text{Paris})$
 $\text{isLocatedIn}(\text{Louvre}, \text{Paris})$

1. Assuming OWA, $T \models \text{paintedBy}(\text{MonaLisa}, \text{LeonardoDaVinci})$ returns *True*.
2. Assuming OWA, $T \models \text{Person}(\text{Bob})$ returns *Idk*.

3. Assuming OWA, $T \models \text{hasVisited}(\text{Bill}, \text{EiffelTower})$ returns *True*.
4. Assuming OWA, $T \models \text{hasTraveledTo}(\text{Bill}, \text{Paris})$ returns *True*.
5. Assuming OWA, $T \models \text{hasTraveledTo}(\text{Bob}, \text{Paris})$ returns *True*.
6. Assuming OWA, $T \models \text{Person}(\text{LeonardoDaVinci})$ returns *False*.
7. Assuming CWA, $T \models \text{isLocatedIn}(\text{Louvre}, \text{Paris})$ returns *True*.
8. Assuming CWA, $T \models \text{exhibitedIn}(\text{MonaLisa}, \text{Louvre})$ returns *False*.
9. Assuming CWA, $T \models \text{isLocatedIn}(\text{MonaLisa}, \text{Paris})$ returns *Idk*.
10. Assuming CWA, $T \models \text{isFriendOf}(\text{Alice}, \text{Bill})$ returns *True*.
11. Assuming CWA, $T \models \text{hasVisited}(\text{Bob}, \text{Louvre})$ returns *True*.
12. Assuming CWA, $T \models \text{traveledTogether}(\text{Bob}, \text{Bill})$ returns *False*.
13. $T \models \text{Person}$ result in $\{\text{Alice}, \text{Bob}, \text{Bill}\}$.
14. $T \models \text{City}$ result in $\{\text{Paris}\}$.
15. $T \models \text{Location}$ result in $\{\text{Paris}, \text{Louvre}\}$.
16. $T \models \text{hasVisited}$ result in $\{\text{Bill}, \text{Bob}\}$.

Exercise 1.9 Considering the following LoE theory T , indicate which of the following statements are true.

VacuumCleaner(VC#1)
VacuumCleaner(VC#2)
Appliance(VC#2)
Appliance(VC#3)
consumedEnergy(VC#1, 120W)
consumedEnergy(VC#2, 1000W)
Cordless(VC#1)
Electric(VC#2)
belongsTo(Mary#1, VC#1)
Person(Mary#1)

1. Assuming CWA, $\{\text{VC\#1}, \text{VC\#2}\}$ are the only entities which are “*VacuumCleaner*”.
2. Assuming OWA, it is unknown whether VC#1 is an “*Appliance*”.
3. Assuming CWA, it is known that VC#3 is not an “*VacuumCleaner*”.
4. Assuming OWA, VC#1 uses a “*Battery*”.
5. Assuming OWA, VC#1 is not “*Elettrico*”.
6. Assuming OWA, Mary#1 is not an “*VacuumCleaner*”.
7. Assuming OWA, Mary#1 is an “*VacuumCleaner*”.

Exercise 1.10 Considering the following LoE theory T , indicate which of the following statements are true.

Person(Mario#1)
Student(Mario#1)
Student(Sara#1)
UniversityCourse(Logic#1)
UniversityCourse(Networking#1)
isAttendingCourse(Mario#1, Logic#1)
isAttendingCourse(Sara#1, Networking#1)
numberOfExamsPassed(Sara#1, 1)
hasMark(Sara#1, 23)

1. Assuming OWA, we know that there are no other entities other than Mario#1 that are in the etype *Person*.
2. Assuming CWA, we know that Sara#1 has passed the exam for the course *Networking*#1.
3. Assuming OWA, we do not know if the only mark that Sara#1 has obtained is 23.
4. Assuming CWA, Mario#1 is not attending *Networking*#1.
5. Assuming CWA, Sara#1 is not a *Person*.
6. Assuming OWA, we know that Sara#1 is not a *Person*.

1.4 Solutions

Solution 1.1

1. True
2. False
3. True

Solution 1.2

1. False
2. True
3. True
4. True
5. False
6. True
7. False
8. False

Solution 1.2

1. True
2. False
3. False
4. True
5. True
6. False
7. True

Solution 1.4

1. True
2. True
3. False
4. True
5. False
6. False

Solution 1.5

1. True
2. False
3. True
4. False

Solution 1.6

1. True
2. True
3. True
4. False
5. False

Solution 1.7

1. False
2. True
3. True
4. False
5. False

Solution 1.2

1. True

2. False
3. True
4. False
5. False

Solution 1.8

1. True
2. False
3. True
4. False
5. True
6. False
7. True
8. False
9. False
10. False
11. True
12. True
13. True
14. True
15. False
16. False

Solution 1.9

1. True
2. True
3. True
4. False
5. False
6. False
7. False

Solution 1.10

1. False
2. False
3. True
4. True
5. True
6. False